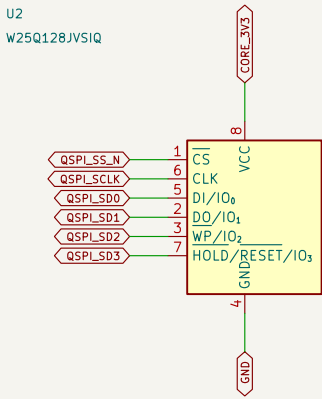
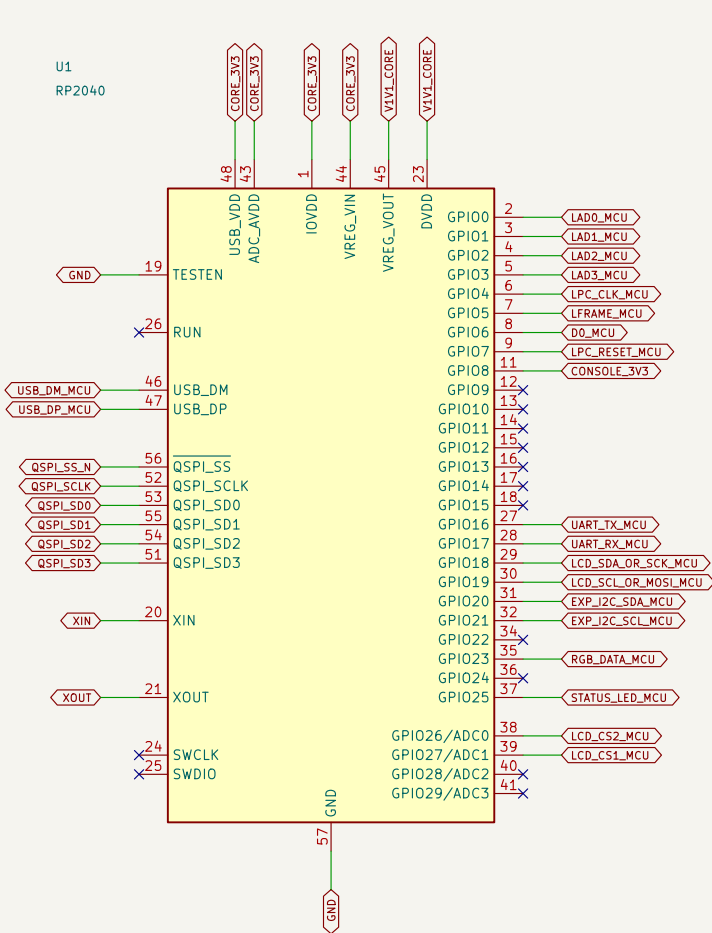


01 RP2040 and flash core

54 observed components; selected completion based on YD2040, photographed copper and recorded specimen readings.
Net membership comes from the native PCB; values, pin roles and deliberate unused contacts come from completion_Contract.json.



RP2040 supply and QSPI wiring follows the reference core.
Flash pin 4 ground / pin 8 supply retain measured anchors.
W25Q128JVSQ is the observed SOIC flash.
Selected values are not all measured original values.

Power selection and measured anchors



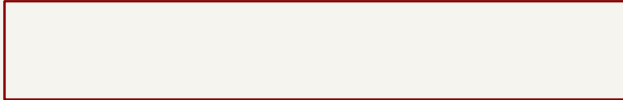
File: MODXO_ApexusNULL_Power.kicad_sch

LPC, USB and test contacts



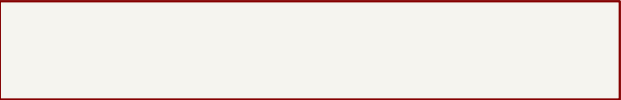
File: MODXO_ApexusNULL_Host_Contacts.kicad_sch

Accessory connector assignments



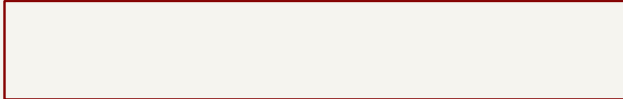
File: MODXO_ApexusNULL_Accessory_Contacts.kicad_sch

Clock, LEDs, BOOT and interface



File: MODXO_ApexusNULL_Peripherals.kicad_sch

Selected passive values and nets



File: MODXO_ApexusNULL_Passives.kicad_sch

Selected completion; measured / inferred provenance retained. Explicit NC per contract.
54 observed footprints; exact assigned PCB nets preserved

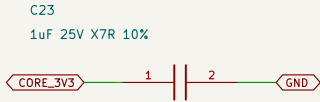
Sheet: /
File: MODXO_ApexusNULL.kicad_sch

Title: Lite V1 – Core reference mappings

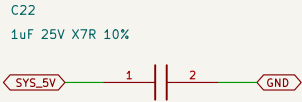
Size: A3	Date: 2026-09-23	Rev: COMPLETE 0.1
KiCad E.D.A. 10.0.6		Id: 1/6

02 Power selection and measured anchors

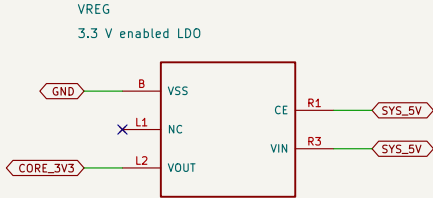
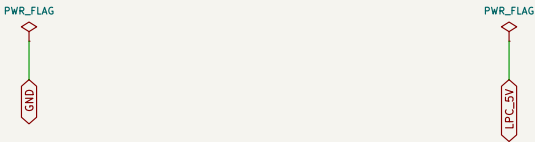
Observed power components retain native pad IDs; inferred manufacturer functions are shown as aliases.
Recorded continuity groups are preserved. Selected part identity and nominal voltages do not constitute new measurements.



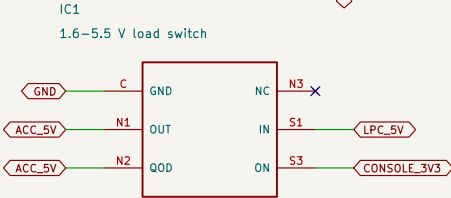
C23.1 = measured E / flash supply.
C23.2 = measured A / ground.
Selected capacitance: see component value.



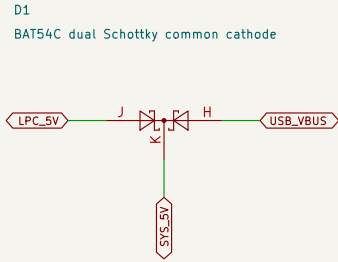
C22.1 = measured F / diode common cathode group.
C22.2 = measured D / ground.
Input/enable arrangement is preserved.



Measured VREG; B ground; L2 flash supply;
R1 and R3 diode-output group.
Function aliases use the selected LDO pinout.
Declared unused BP/NC terminal stays open.



Measured IC1: C ground; S1 joins D1.J.
TPS22919 identity/orientation is inferred.
Enable, output and QOD connections are completion
choices recorded in the final contract.



D1 observed H/J/K remain unchanged.
Selected BAT54C common-cathode topology.
Recorded diode-mode drops are in-circuit evidence.
Pin names are hidden to keep the diode glyph legible.

User reports the original specimen works and boots PrometheOS; all leads are soldered down.
This reconstruction combines measured memberships, optical deductions and explicit engineering selections.

Selected completion; measured / inferred provenance retained. Explicit NC per contract.
54 observed footprints; exact assigned PCB nets preserved

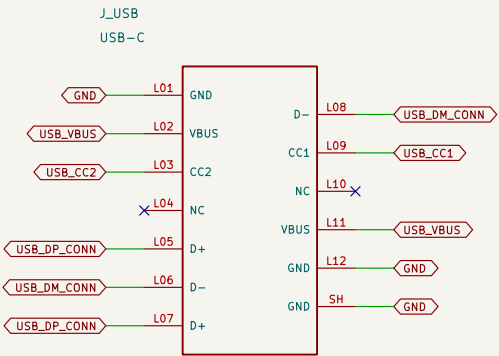
Sheet: /Power selection and measured anchors/
File: MODX0_ApexusNULL_Power.kicad_sch

Title: Lite V1 – Measured power

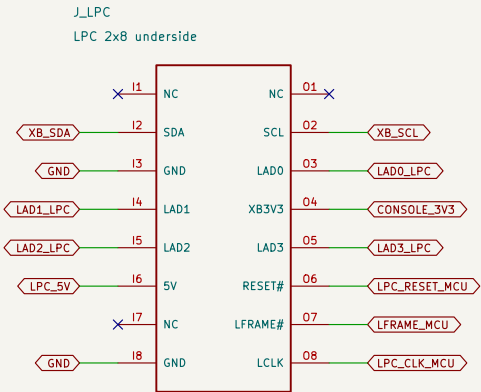
Size: A3	Date: 2026-09-23	Rev: COMPLETE 0.1
KiCad E.D.A. 10.0.6		Id: 2/6

03 LPC, USB and test contacts

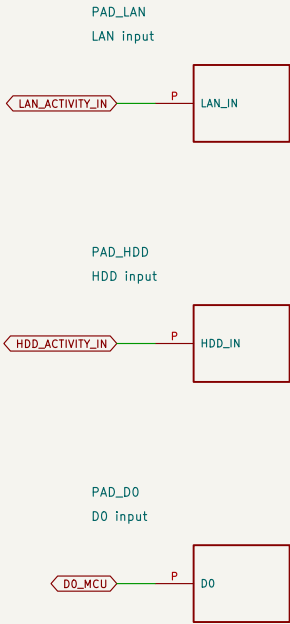
USB neutral tail IDs L01–L12 are mapped to selected TYPEC–304 functions; no pad–ID substitution is hidden.
LPC row observations map to the physical console pinout documented by the completion contract.



USB L01–L12 are left–to–right with USB north.
L05/L07 = D+; L06/L08 = D–.
L03 and L09 = independent CC pulldowns.
L04/L10 SBU contacts are deliberately unused.
Shell grounding is selected for this completion.



LPC I/O means inner / outer physical row.
See the completion contract for console pin aliases.
Unused contacts have explicit NC markers.
Existing measured and optical memberships
are retained except documented corrected readings.



Earlier R8–to–USB–L05 alignment was superseded by the manufacturer pinout and full–YD data–pair topology.
That correction is documented in completion_core.json; it is not presented as a new continuity reading.

Selected completion; measured / inferred provenance retained. Explicit NC per contract.
54 observed footprints; exact assigned PCB nets preserved

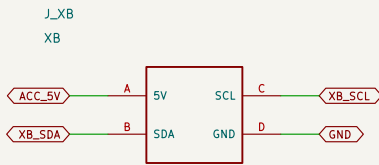
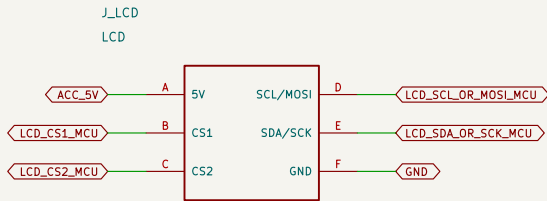
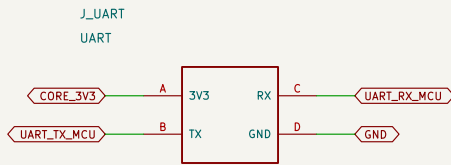
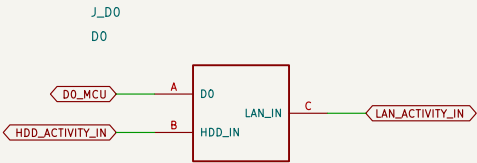
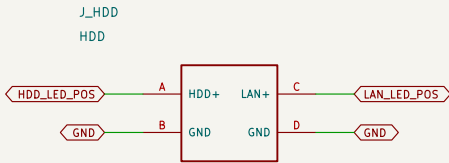
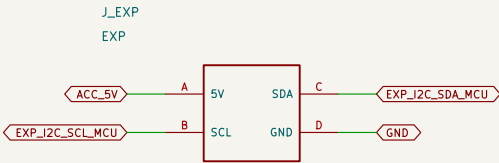
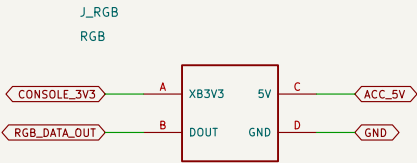
Sheet: /LPC, USB and test contacts/
File: MODX0_ApexusNULL_Host_Contacts.kicad_sch

Title: Lite V1 – Host contacts

Size: A3 Date: 2026–09–23 Rev: **COMPLETE 0.1**
KiCad E.D.A. 10.0.6 Id: 3/6

04 Accessory connector assignments

Physical connector counts, positions and observation IDs are retained.
Function and supply assignments are explicit completion selections; mating contact numbering is documented in the contract.



XBI2C and EXPI2C remain distinct connector functions. Four RGB contacts are retained.
Selected connector family and pad numbering must agree with the final footprint qualification.

Selected completion; measured / inferred provenance retained. Explicit NC per contract.
54 observed footprints; exact assigned PCB nets preserved

Sheet: /Accessory connector assignments/
File: MODX0_ApexusNULL_Accessory_Contacts.kicad_sch

Title: Lite V1 – Accessory contacts

Size: A3 Date: 2026–09–23

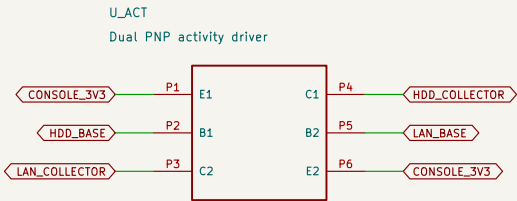
KiCad E.D.A. 10.0.6

Rev: COMPLETE 0.1

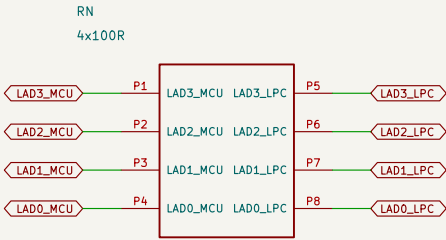
Id: 4/6

05 Clock, LEDs, BOOT and interface

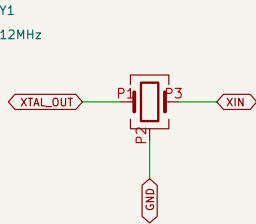
Selected component functions use existing physical observation IDs.
Clock, BOOT and LED topology is completed without adding fitted components; source assumptions stay documented.



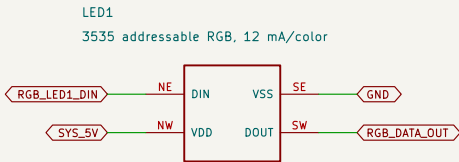
Selected activity circuit.
Native P1–P6 roles are aliases from the completion contract; original lane interpretations and corrections remain in the evidence register.



Four isolated 100-ohm branches.
Native pad functions are shown on the symbol.
Manufacturer opposing-pad pairing is documented in the selected resistor-array specification.



Y1 = 12MHz, selected CL10pF crystal.
C12/C13 = 15pF; R5 = 1k series drive.
C13:B-to-Y1.P3 is retained optical evidence.
Native P1–P4 remain corner observation IDs.



BOOT completion directly grounds QSPI chip select. Hold BOOT before applying USB power.
No additional CS pull-up or BOOT series resistor has been inserted into the observed 54-part population.



Selected completion; measured / inferred provenance retained. Explicit NC per contract.
54 observed footprints; exact assigned PCB nets preserved

Sheet: /Clock, LEDs, BOOT and interface/
File: MODX0_ApexusNULL_Peripherals.kicad_sch

Title: Lite V1 – Observed peripherals

Size: A3	Date: 2026-09-23	Rev: COMPLETE 0.1
KiCad E.D.A. 10.0.6		Id: 5/6

06 Selected passive values and nets

Values are selected for the completed circuit; readable component markings remain separate evidence.
R1 is marked 102 (1k), RN is marked 101 (four 100-ohm elements); capacitor values are engineering selections.

C16
100nF 50V X7R 10%



R10
5.1k 1% 0.1W



R8
27R 1% 0.1W



R7
27R 1% 0.1W



R4
5.1k 1% 0.1W



P_N1
1uF 25V X7R 10%



P_N2
100nF 50V X7R 10%



C9
100nF 50V X7R 10%



C13
15pF 50V C0G 5%



C12
15pF 50V C0G 5%



R5
1k 1% 0.1W



C5
100nF 50V X7R 10%



C7
100nF 50V X7R 10%



C6
100nF 50V X7R 10%



C11
100nF 50V X7R 10%



C4
2.2uF 10V X7R 10%



C2
2.2uF 10V X7R 10%



C8
100nF 50V X7R 10%



C10
100nF 50V X7R 10%



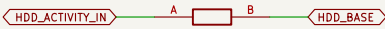
C3
100nF 50V X7R 10%



R1
1k 1% 0.1W



R9
4.7k 1% 0.1W



R3
470 1% 0.1W



R2
470 1% 0.1W



R11
4.7k 1% 0.1W



C15
2.2uF 10V X7R 10%



C14
100nF 50V X7R 10%



C19
1uF 25V X7R 10%



R6
330 1% 0.1W



Selected completion; measured / inferred provenance retained. Explicit NC per contract.
54 observed footprints; exact assigned PCB nets preserved

Sheet: /Selected passive values and nets/
File: MODX0_ApexusNULL_Passives.kicad_sch

Title: Lite V1 – Observed passives

Size: A3 Date: 2026-09-23

KiCad E.D.A. 10.0.6

Rev: COMPLETE 0.1

Id: 6/6